

Seaborn:

* which makes statistical visualization even, more attractive and give more information.
* Seaborn is a python data library built on top of matplotlib, it provides a high level interface for creating beautiful and informative statistical graphs (professional-looking graph with less code).

* built in statistical plots

matplotlib (X) $\Rightarrow$ Interview

Basic visualization

more customization required

more coding

good for simple plots

Base plotting library.

Seaborn :
statistical visualization
attractive by default

less Coding
Good for Complex and professional plots
built on top of matplotlib.

Install Seaborn:
pip install seaborn

```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd.
```

Built in dataset: $\rightarrow$ seaborn provides several built in datasets

ex: tips loops $\rightarrow$ use max + ymn

1912
penguin

diamonds

+ lights

titanic

df = sns.load_dataset("tips")
(df.head())

```

df = sns.load_data()
print(df.head())

```

Dataset information view (-)

$$\Rightarrow T_{alt} = 1150^\circ C$$

shape:

dt.shape()

Column names check():

def. columns

statistical Summary

dt. describe ()

Theme :

```
sns.set_theme(style="darkgrid")
sns.set_theme(style="whitegrid")
sns.set_theme(style="white")
sns.set_theme(style="ticks")
```

- Qn:
1. install seaborn and import it
 2. load the iris dataset
 3. display: data inspection
 4. ~~use~~ apply different seaborn themes and observe the changes

lineplot seaborn syntax:

```
sns.lineplot(data=df, x="column 1",
              y="column 2")
```

import seaborn as sns

import matplotlib.pyplot as plt

```
df = sns.load_dataset("tips")
```

```
sns.lineplot(data=df, x="size", y="total-bill")
```

```
plt.show()
```

Bar plot: (avg)

```
sns.barplot(data=df, x="day", y="total-bill")
```

Count plot: (frequency)

```
sns.countplot(data=df, x="day")
```

Counts the number of Occurrence of each category no numerical Column is required

Barplot: Average Countplot: Frequency

Scatter plot:

```
sns.scatterplot(data=df, x="total-bill", y="tip")
```


Hue:

* powerful feature in Seaborn.

* It separates the categories using different colours.

~~Plot~~

`sns.scatterplot(data=df, x="total_bill", y="tip", hue="sex")`

Add style

smoker

`sns.scatterplot(data=df, x="total_bill", y="tip", style="smoker")`

size:

`sns.scatterplot(data=df, x="total_bill", y="tip", size="size")`

How many positive plots: customer analytics

Countplot

Averaging per day

Barplot

Sales growth over month

lineplot

~~vs~~: Review length vs sentiment score

scatter

↓
vs. → scatterplot

1. load another dataset?
2. Create a line plot and show any two columns plotted
3. Create a barplot
4. Create a countplot (x)
5. Create a scatter plot (vs)
6. hue
7. style
8. size